

Selective defect creation in 2D hexagonal boron nitride via ultra-low energy Ar ion irradiation

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In recent years, point defects in hexagonal boron nitride (hBN) have received increasing attention for being excellent hosts for single photon emitters (SPE) at room temperature [1,2]. SPEs are a growing research area with potential applications in quantum communication and information technology, quantum simulations, as well as photonics [3], which introduces the question of whether defect creation in hBN can be controlled. So far, no conclusive results have been reported correlating the exact defect structure and the quantum emission [4,5]. According to simulations using time dependent density functional theory (TDDFT), an Ar^{+1} ion in a head-on collision with nitrogen requires 34.5 eV to successfully knock it out from the hBN lattice, and 45.5 eV to successfully knock out a boron atom, indicating the possibility of selective defect creation in hBN. In this experimental study, freestanding monolayer hBN samples were prepared using the electrochemical delamination method and subjected to 150 eV Ar^{+1} ion irradiation. Using our interconnected ultra-high vacuum system [6], the irradiation effects were directly characterized via scanning transmission electron microscopy (STEM) without exposing the samples to ambient conditions. With the help of a semi-automated STEM image acquisition tool [7], built into our microscope's software, thousands of atomic resolution images of the hBN samples were acquired before and after the irradiation. The acquired images were then analyzed by a semi-autonomous convolutional neural network (CNN) that can identify the hBN lattice, vacancies, and substitutions, giving us an understanding of the defect types and their distributions. By irradiating with an energy higher than the defect creation energy for both boron and nitrogen, we find a surprising result where the ratio between the defects are not equal but have a significant difference that cannot be explained by simulations. From the analysis, the as-prepared samples have an intrinsic defect density of 0.023 nm^{-2} with a roughly 1:1 ratio in B:N single vacancies, but ca. 40% of the boron single vacancies were found to be filled by Si atoms. After irradiation, we found a defect density of 0.243 nm^{-2} with a roughly 85:15 ratio in B:N single vacancies and respective 3:1 and 6:1 ratio of single-to-double and single-to-triple vacancies. We also observe that many vacancies have been filled by Si and C

atoms. Overall, we show a high defect creation probability at this ion energy and demonstrate that such irradiation already provides selectivity in the defect types created.

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- [2] Grosso et al., *Nature Communications* **8**, 705 (2017).
- [3] Liu et al., *Nano Materials Science* **3**, 291–312 (2021).
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- [7] Mittelberger et al., *Microscopy and Microanalysis* **23**, 809–817 (2017).